Project Folder Structure and Workflow Design

Port Authority Capstone

1. Purpose of This Structure

Before starting the analysis phase, we created a structured project environment to keep our work organized, reproducible, and collaborative. Since this is a group project with multiple phases (data audit, EDA, cleaning, integration, modeling), we needed a clean system that prevents file confusion, separates raw and processed data, supports automation, makes collaboration easier, and keeps the final deliverables professional.

This structure ensures that all team members work within a consistent framework.

2. Folder Structure Overview

The main project folder is: `PortAuthority_Capstone`

Inside it, we created the following folders:

1. data/

Used to store all datasets. This separation ensures we never overwrite original data.

- `raw/` – Original datasets provided by the company
- `processed/` – Processed Datasets
- `cleaned/` – Cleaned Datasets
- `integrated/` – Final merged dataset ready for modeling

2. **notebooks/**

Contains all Jupyter notebooks used for analysis. Each notebook corresponds to a specific phase, keeping analytical work modular and easy to follow:

- Data audit
- EDA (Exploratory Data Analysis)
- Data cleaning
- Data integration
- Sampling and final dataset

3. **scripts/**

Contains reusable Python functions. Instead of writing repeated code inside notebooks, we separate reusable logic here. This improves automation and maintainability.

4. **charts/**

Stores all exported visualizations. Saving charts separately ensures consistency in reports and presentations. Subfolders include:

- `eda/`
- `cleaning/`
- `integration/`
- `final_dataset/`

5. **reports/**

Contains all documentation deliverables. This keeps documentation separate from code and data:

- Group reports
- Individual member reports
- Presentation files

6. **documentation/**

Used for internal notes to help maintain transparency and support presentation preparation:

- Audit observations
- Cleaning decisions
- Integration logic
- Assumptions made during analysis

3. Why This Structure Is Necessary

By setting up the structure before starting technical work, we ensure the project remains organized and manageable throughout the semester. This structure supports:

- **Clear separation of responsibilities** among group members
- **Reproducible analysis** ensuring any team member can run the code
- **Automation** of data processing workflows
- **Professional presentation standards** for deliverables
- **Scalability** for future modeling phases

4. Conclusion

The project structure was intentionally designed to reflect real-world analytics workflows. It allows us to move systematically from raw data to a final modeling-ready dataset while maintaining clarity, collaboration, and version control.This foundation will support all future phases of the capstone project.